

BIKE SHOP ANALYSIS



Data Analytics Project
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Tools: Excel, SQL, Power Bi

Project Overview



- ~Analyze bike shop performance.
- ~Identify trends in revenue, profit, riders, and seasonal performance.
- ~Build an interactive Power Bi dashboard.

Tools Used



Excel

~Data cleaning &
initial exploration



Sql
server

~Data Analysis



Power Bi

~Dashboard creation

Dataset



Columns:

Date, Revenue, Profit,
Riders, Season, Hour, Year.

Data source:

Kaggle-Bike Sharing Dataset

Dashboard KPIs



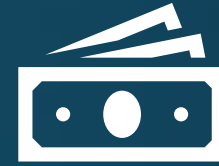
\$ 15M

Sum of revenue



3M

Sum of riders



\$10.45M

Sum of profit



0.45

Profit margin

Key Insights



- ➔ Peak sales occur between 10 Am and 3Am.
- ➔ Revenue & profit Increase during summer months.
- ➔ The majority of riders are registered users.
- ➔ Wednesday & Friday generate higher revenue.

Dashboard Explanation

- ~KPI Cards
- Top level performance summary
- ~Monthly KPI Trend
- Trend of riders, profit & revenue over time.
- ~Revenue by season
- Comparison of revenue across seasons.
- ~Rider Demographic
- Customer segment distribution.
- ~Hourly Sales Heatmap
- Sales pattern by hour & weekday
- ~Year Slicer
- Filter dashboard by year.



Business Recommendations



1. Optimize Peak Hours : Increase marketing & staff during peak hours to maximize revenue.
2. Focus on High-Profit Days: Run special offers on Wednesdays and Fridays to boost sales.
3. Seasonal Promotions: launch campaigns during off- peak seasons to maintain consistent revenue.
4. Loyalty Program: Reward registered riders & encourage casual riders to become registered users.

Challenges Faced

- Data cleaning and preparing.
 - ~Writing DAX measures for KPIs.
- Choosing suitable visualizations.
- Creating relationships between tables.

Conclusion

- This project enhanced my skills in Sql, Excel, and Power Bi by transforming raw data into meaningful insights. The dashboard provides valuable business insights to support better decision-making and optimize operations.